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Electrosurgical device with illumination and smoke evacuation features

Abstract

In an example, an electrosurgical device includes a housing defining an interior bore. The electrosurgical device also includes a shaft telescopically moveable in the interior bore of the housing. The shaft includes an optical waveguide at a distal end of the shaft, and a smoke evacuation channel circumferentially surrounding the optical waveguide at the distal end of the shaft. The electrosurgical device also includes an electrosurgical electrode coupled to the shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a U.S. national phase of International Application No. PCT/IB2020/000946, filed on Nov. 12, 2020, which claims the benefit of U.S. Provisional Patent Application No. 62/976,744, filed on Feb. 14, 2020, and U.S. Provisional Patent Application No. 62/934,512 filed on Nov. 12, 2019, the contents of which are hereby incorporated by reference in their entirety.

FIELD

(1) The present disclosure generally relates to methods and apparatus for conveying electrical energy and, more specifically, to electrosurgical devices and the methods providing for telescopic adjustment of an electrosurgical electrode.

BACKGROUND

(2) Electrosurgery involves applying a radio frequency (RF) electric current (also referred to as electrosurgical energy) to biological tissue to cut, coagulate, or modify the biological tissue during an electrosurgical procedure. Specifically, an electrosurgical generator generates and provides the electric current to an active electrode, which applies the electric current (and, thus, electrical power) to the tissue. The electric current passes through the tissue and returns to the generator via a return electrode (also referred to as a “dispersive electrode”). As the electric current passes through the tissue, an impedance of the tissue converts a portion of the electric current into thermal energy (e.g., via the principles of resistive heating), which increases a temperature of the tissue and induces modifications to the tissue (e.g., cutting, coagulating, ablating, and/or sealing the tissue).

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) The novel features believed characteristic of the illustrative examples are set forth in the appended claims. The illustrative examples, however, as well as a preferred mode of use, further objectives and descriptions thereof, will best be understood by reference to the following detailed description of an illustrative example of the present disclosure when read in conjunction with the accompanying drawings, wherein:

(2) FIG. 1 depicts a simplified block diagram of an electrosurgical system, according to an example.

(3) FIG. 2 depicts a cross-sectional view of an electrosurgical device, according to an example.

(4) FIG. 3 depicts a cross-sectional view of an electrosurgical device, according to an example.

- (5) FIG. 4 depicts a perspective view of an electrosurgical device, according to an example.
- (6) FIG. 5 depicts a distal end of an electrosurgical device, according to an example.
- (7) FIG. 6 depicts a distal end of an electrosurgical device, according to another example.
- (8) FIG. 7A depicts a perspective view of an electrosurgical device, according to another example.
- (9) FIG. 7B depicts a perspective view of a distal portion of the electrosurgical device shown in FIG. 7A, according to an example.
- (10) FIG. 7C depicts shows a side view of the electrosurgical device shown in FIG. 7A, according to an example
- (11) FIG. 8A depicts a perspective view of a distal end portion of a shaft of an electrosurgical device, according to example.
- (12) FIG. 8B depicts a side view of the distal end portion shown in FIG. 8A, according to example.
- (13) FIG. 9A depicts a perspective view of an electrosurgical device, according to another example.
- (14) FIG. 9B depicts a perspective view of a distal end of a shaft of the electrosurgical device shown in FIG. 9A, according to an example.
- (15) FIG. 9C depicts a side view of the distal end of the shaft shown in FIG. 9B, according to an example.
- (16) FIG. 9D depicts a cross-sectional view of the electrosurgical device shown in FIG. 9A, according to an example.
- (17) FIG. 9E depicts an exploded view of the electrosurgical device shown in FIG. 9A, according to an example.
- (18) FIG. 10A depicts a perspective view of an electrosurgical device 112, according to another example.
- (19) FIG. 10B depicts a perspective view of a distal end of a shaft of the electrosurgical device shown in FIG. 10A, according to an example.
- (20) FIG. 10C depicts a side view of a distal end of the shaft of the electrosurgical device shown in FIG. 10B, according to an example.
- (21) FIG. 10D depicts another side view of a distal end of the shaft of the electrosurgical device shown in FIG. 10B, according to an example.
- (22) FIG. 10E depicts a light-emitting diode printed circuit board of the electrosurgical device shown in FIG. 10A, according to an example.
- (23) FIG. 10F depicts a cross-sectional view of the electrosurgical device shown in FIG. 10A, according to an example.
- (24) FIG. 11 depicts light-emitting diode printed circuit boards for the electrosurgical device shown in FIG. 10A, according to examples.

DETAILED DESCRIPTION

- (25) Disclosed examples will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all of the disclosed examples are shown. Indeed, several different examples may be described and should not be construed as limited to the examples set forth herein. Rather, these examples are described so that this disclosure will be thorough and complete and will fully convey the scope of the disclosure to those skilled in the art.
- (26) By the term “approximately” or “substantially” with reference to amounts or measurement values described herein, it is meant that the recited characteristic, parameter, or value need not be achieved exactly, but that deviations or variations, including for example, tolerances, measurement error, measurement accuracy limitations and other factors known to those of skill in the art, may occur in amounts that do not preclude the effect the characteristic was intended to provide.
- (27) As noted above, an electrosurgical device can use electrical energy supplied by an electrosurgical generator to apply electrosurgical energy from an electrosurgical electrode to a tissue. As such, the electrosurgical device generally includes a housing in which one or more conductors are disposed for supplying the electrosurgical energy to the electrosurgical electrode. Some electrosurgical devices include a shaft that is telescopically adjustable relative to the housing.

This can facilitate adjusting a length of the electrosurgical device to treat differently sized and/or shaped target tissues.

(28) Referring now to FIG. 1, an electrosurgical system **100** is shown according to an example. As shown in FIG. 1, the electrosurgical system **100** includes an electrosurgical generator **110** and an electrosurgical device **112**. In general, the electrosurgical generator **110** can generate electrosurgical energy that is suitable for performing electrosurgery on a patient. For instance, the electrosurgical generator **110** can include a power converter circuit **114** that can convert a grid power to electrosurgical energy such as, for example, a radio frequency (RF) output power. As an example, the power converter circuit **114** can include one or more electrical components (e.g., one or more transformers) that can control a voltage, a current, and/or a frequency of the electrosurgical energy.

(29) Within examples, the electrosurgical generator **110** can include a user interface **116** that can receive one or more inputs from a user and/or provide one or more outputs to the user. As examples, the user interface **116** can include one or more buttons, one or more switches, one or more dials, one or more keypads, one or more touchscreens, and/or one or more display screens.

(30) In an example, the user interface **116** can be operable to select a mode of operation from among a plurality of modes of operation for the electrosurgical generator **110**. As examples, the modes of operation can include a cutting mode, a coagulating mode, an ablating mode, and/or a sealing mode. Combinations of these waveforms can also be formed to create blended modes. In one implementation, the modes of operation can correspond to respective waveforms for the electrosurgical energy. As such, in this implementation, the electrosurgical generator **110** can generate the electrosurgical energy with a waveform selected from a plurality of waveforms based, at least in part, on the mode of operation selected using the user interface **116**.

(31) The electrosurgical generator **110** can also include one or more sensors **118** that can sense one or more conditions related to the electrosurgical energy and/or the target tissue. As examples, the sensor(s) **118** can include one or more current sensors, one or more voltage sensors, one or more temperature sensors, and/or one or more bioimpedance sensors. Within examples, the electrosurgical generator **110** can additionally or alternatively generate the electrosurgical energy with an amount of electrosurgical energy (e.g., an electrical power) and/or a waveform selected from among the plurality of waveforms based on one or more parameters related to the condition(s) sensed by the sensor(s) **118**.

(32) In one example, the electrosurgical energy can have a frequency that is greater than approximately 100 kilohertz (kHz) to reduce (or avoid) stimulating a muscle and/or a nerve near the target tissue. In another example, the electrosurgical energy can have a frequency that is between approximately 300 kHz and approximately 500 kHz.

(33) In FIG. 1, the electrosurgical generator **110** also includes a connector **120** that can facilitate coupling the electrosurgical generator **110** to the electrosurgical device **112**. For example, the electrosurgical device **112** can include a power cord **122** having a plug, which can be coupled to a socket of the connector **120** of the electrosurgical generator **110**. In this arrangement, the electrosurgical generator **110** can supply the electrosurgical energy to the electrosurgical device **112** via the coupling between the connector **120** of the electrosurgical generator **110** and the power cord **122** of the electrosurgical device **112**.

(34) As shown in FIG. 1, the electrosurgical device **112** can include a housing **124** defining an interior bore **125**, a shaft **126** extending in a distal direction from the housing **124**, and an electrosurgical electrode **128** coupled to the shaft **126**. In general, the housing **124** can be configured to facilitate a user gripping and manipulating the electrosurgical device **112** while performing electrosurgery. For example, the housing **124** can have a shape and/or a size that can facilitate a user performing electrosurgery by manipulating the electrosurgical device **112** using a single hand. In one implementation, the housing **124** can have a shape and/or a size that facilitates the user holding the electrosurgical device **112** in a writing utensil gripping manner (e.g., the electrosurgical device **112** can be an electrosurgical pencil).

(35) Additionally, for example, the housing **124** can be constructed from one or more materials that are electrical insulators (e.g., a plastic material). This can facilitate insulating the user from the electrosurgical energy flowing through the electrosurgical device **112** while performing the electrosurgery.

(36) In some implementations, the shaft **126** can be fixedly coupled to the housing **124**. In other implementations, the shaft **126** can be telescopically moveable relative to the housing **124**. For example, the shaft **126** can be telescopically moveable in an interior bore defined by the housing **124** to extend the shaft **126** in the distal direction and retract the shaft **126** in a proximal direction relative to the housing **124** (e.g., movable along a longitudinal axis of the electrosurgical device **112**). As noted above, the electrosurgical electrode **128** is coupled to the shaft **126** and, thus, the electrosurgical electrode **128** moves together with the shaft **126** relative to the housing **124**. This can provide for adjusting a length of the electrosurgical device **112**, which can facilitate performing electrosurgery at a plurality of different depths within tissue (e.g., due to different anatomical shapes and/or sizes of patients) and/or at a plurality of different angles.

(37) In some examples, the shaft **126** can additionally or alternatively be rotatable about an axis of rotation that is parallel to the longitudinal axis of the electrosurgical device **112**. In another example, the electrosurgical electrode **128** can be additionally or alternatively rotatable relative to the shaft **126**. Rotating the shaft **126** and/or the electrosurgical electrode **128** relative to the housing **124** can facilitate adjusting an angle of the electrosurgical electrode **128** relative to one or more user input device(s) **130** of the electrosurgical device **112**. In this arrangement, a user can comfortably grip the housing **124** in a position in which their fingers can comfortably operate the user input device(s) **130** while the electrosurgical electrode **128** is set at a rotational position selected from among a plurality of rotational positions relative to the housing **124** based on, for example, a location, a size, and/or a shape of a surgical site in which the user is operating.

(38) The user input device(s) **130** can select between the modes of operation of the electrosurgical device **112** and/or the electrosurgical generator **110**. For instance, in one implementation, the user input device(s) **130** can be configured to select between a cutting mode of operation and a coagulation mode of operation. Responsive to actuation of the user input device(s) **130** of the electrosurgical device **112**, the electrosurgical device **112** can (i) receive the electrosurgical energy with a level of power and/or a waveform corresponding to the mode of operation selected via the user input device(s) **130** and (ii) supply the electrosurgical energy to the electrosurgical electrode **128**.

(39) In FIG. **1**, the electrosurgical device **112** includes a plurality of electrical components that facilitate supplying the electrosurgical energy, which the electrosurgical device **112** receives from the electrosurgical generator **110**, to the electrosurgical electrode **128**. For example, the electrosurgical device **112** can include a printed circuit board **132** (e.g., a flexible printed circuit board), a housing conductor **134**, one or more conductive leads **136**, and/or a receptacle **137** that can provide a circuit for conducting the electrosurgical energy from the power cord **122** to the electrosurgical electrode **128**. One or more of the electrical components can be positioned in the inner bore defined by the housing **124**.

(40) Within examples, the user input device(s) **130** can include one or more buttons on an exterior surface of the housing **124**. Each button of the user input device(s) **130** can be operable to actuate a respective one of a plurality of switches **138** of the printed circuit board **132**. In general, the switches **138** and/or the printed circuit board **132** are operable to control a supply of the electrosurgical energy from the electrosurgical generator **110** to the electrosurgical electrode **128**. For instance, in one implementation, when each button is operated (e.g., depressed), the respective switch **138** associated with the button can be actuated to cause the printed circuit board **132** to transmit a signal to the electrosurgical generator **110** and cause the electrosurgical generator **110** to responsively supply the electrosurgical energy with a level of power and/or a waveform corresponding to a mode of operation associated with the button. In another implementation,

operating the button and thereby actuating the respective switch **138** associated with the button can close the switch **138** to complete a circuit to the electrosurgical generator **110** to cause the electrosurgical generator **110** to responsively supply the electrosurgical energy with a level of power and/or a waveform corresponding to a mode of operation associated with the button. In some examples of this implementation, the printed circuit board **132** can be omitted.

(41) In both example implementations, the electrosurgical energy supplied by the electrosurgical generator **110** can be supplied from (i) the power cord **122**, the printed circuit board **132**, and/or the switches **138** to (ii) the electrosurgical electrode **128** by the housing conductor **134** and the conductive lead(s) **136**. As such, as shown in FIG. **1**, the printed circuit board **132** can be coupled to the power cord **122**, the housing conductor **134** can be coupled to the printed circuit board **132** and the conductive lead(s) **136**, and the conductive lead(s) **136** can be coupled to the electrosurgical electrode **128** (e.g., via the receptacle **137**). In this arrangement, the housing conductor **134** can conduct the electrosurgical energy (supplied to the housing conductor **134** via the printed circuit board **132**) to the conductive lead(s) **136**, and the conductive lead(s) **136** and the receptacle **137** can conduct the electrosurgical energy to the electrosurgical electrode **128**.

(42) In general, the housing conductor **134** can include one or more conductive elements that provide an electrically conductive bus for supplying the electrosurgical energy to the electrosurgical electrode **128**. In one example, the housing conductor **134** can be formed in a helical shape. In this arrangement, the housing conductor **134** can be compressible and expandable such that the housing conductor **134** can accommodate the shaft **126** telescopically moving into and/or out of the housing **124** to retract and/or extend, respectively, the electrosurgical electrode **128** relative to the housing **124**. In another example, the conductive lead(s) **136** can include one or more wires. In another example, the conductive lead(s) **136** can include one or more conductive traces formed by, for instance, screen printing, sputtering, electroplating, conductive paint, and/or laser ablation.

(43) Within examples, the conductive lead(s) **136** can extend from the housing conductor **134** to the electrosurgical electrode **128**. In one example, the conductive lead(s) **136** can include one or more wires. In another example, the conductive lead(s) **136** can include one or more conductive traces formed by, for instance, screen printing, sputtering, electroplating, conductive paint, and/or laser ablation. The conductive lead(s) **136** can be disposed in an internal conduit of the shaft **126** and an exterior surface of the shaft **126** can be formed of an electrically insulating material. This can help reduce (or prevent) loss of the electrosurgical energy prior to the electrosurgical electrode **128**.

(44) The receptacle **137** can couple the electrosurgical electrode **128** to the electrosurgical device **112**. As an example, the receptacle **137** and the electrosurgical electrode **128** can be configured to couple to each other by friction-fit. Accordingly, the receptacle **137** and the electrosurgical electrode **128** can have respective sizes and/or respective shapes that provide for a friction-fit coupling between the receptacle **137** and the electrosurgical electrode **128** when the electrosurgical electrode **128** is inserted in the receptacle **137**. This can allow for the electrosurgical electrode **128** to be releasably coupled to the electrosurgical device **112**, which can facilitate an interchangeability of a plurality of the electrosurgical electrodes **128** with the electrosurgical device **112**. The receptacle **137** and electrosurgical electrode **128** can be mechanically keyed to ensure the correct electrical connections are made. In other examples, the electrosurgical electrode **128** can be coupled to the receptacle **137** by another type of releasable coupling (e.g., a threaded coupling) or a non-releasable coupling (e.g., via welding and/or soldering).

(45) Within examples, the receptacle **137** can also include a conductor that can electrically couple the electrosurgical electrode **128** to the electrosurgical energy supplied to the electrosurgical device **112** by the electrosurgical generator **110**. For instance, the receptacle **137** can be electrically coupled to the conductive lead(s) **136** (e.g., by a conductive material).

(46) As shown in FIG. **1**, the electrosurgical device **112** can additionally include a light source **140** that is configured to emit light. In the example of FIG. **1**, the light source **140** can be optically

coupled to an optical waveguide **142**, which is configured to receive the light emitted by the light source **140** and transmit the light in a distal direction toward a surgical site to illuminate the surgical site while performing electrosurgery using the electrosurgical electrode **128**. Within examples, the optical waveguide **142** can transmit the light in the distal direction via total internal reflection. For instance, the optical waveguide can include a cladding and/or an air gap on an exterior surface of the optical waveguide **142**. In some implementations, the optical waveguide **142** can be formed as a single, monolithic structure.

(47) In another example, the electrosurgical device **112** can omit the optical waveguide **142** and instead emit the light from the light source **140** directly to the surgical field without transmitting the light through the optical waveguide **142**. In another example, the electrosurgical device **112** can include one or more optical fibers in addition or alternative to the optical waveguide **142**, and the optical fiber(s) can transmit the light emitted by the light source **140** in the distal direction toward the surgical site.

(48) In FIG. **1**, the light source **140** is coupled to the shaft **126**. As such, the light source **140** can also move telescopically with the shaft **126** relative to the housing **124**. However, in other examples, the light source **140** can be in the interior bore of the housing **124** and/or coupled to an exterior surface of the housing **124**. As examples, the light source **140** can include one or more light emitting diodes (LEDs), organic light emitting diodes (OLEDs), optical fibers, non-fiber optic waveguides, and/or lenses.

(49) The optical waveguide **142** can be at a distal end of the shaft **126**. In some examples, the electrosurgical electrode **128** can extend from a central portion of the optical waveguide **142**. As such, the optical waveguide **142** can circumferentially surround the electrosurgical electrode **128** to emit the light distally around all sides of the electrosurgical electrode **128**. This can help to mitigate shadows and provide greater uniformity of illumination in all rotational alignments of the shaft **126** relative to the housing **124** and/or the electrosurgical device **112** relative to the target tissue.

(50) In implementations that include the light source **140**, the user input device(s) **130**, the printed circuit board **132**, the switches **138**, the housing conductor **134**, and/or the conductive lead(s) **136** can additionally supply an electrical power from a direct current (DC) power source **144** to the light source **140**. In one example, the DC power source **144** can include a battery disposed in the housing **124** and/or the plug of the power cord **122**. Although the electrosurgical device **112** includes the DC power source **144** in FIG. **1**, the DC power source **144** can be separate and distinct from the electrosurgical device **112** in other examples. For instance, in another example, the electrosurgical generator **110** can include the DC power source **144**.

(51) Additionally, in implementations that include the light source **140**, the user input device(s) **130** can be operable to cause the light source **140** to emit the light. In one example, the user input device(s) **130** can include a button that independently controls the light source **140** separate from the button(s) that control the electrosurgical operational modes of the electrosurgical device **112**. In another example, the user input device(s) **130** and the printed circuit board **132** can be configured such that operation of the button(s) that control the electrosurgical operational mode simultaneously control operation of the light source **140** (e.g., the light source **140** can be automatically actuated to emit light when a button is operated to apply the electrosurgical energy at the electrosurgical electrode **128**).

(52) As shown in FIG. **1**, responsive to operation of the user input device(s) **130** to actuate the light source **140**, the DC power source **144** can supply the electrical power (e.g., a DC voltage) to the light source **140** via the printed circuit board **132**, the housing conductor **134**, and/or the conductive lead(s) **136**. In this implementation, one or more of the conductive elements of the housing conductor **134** can be configured to supply the electrical power from the DC power source **144** to the light source **140** and/or return the electrical power from the light source **140** to the DC power source **144**. Accordingly, the housing conductor **134** can additionally or alternatively assist in providing electrical communication between the DC power source **144** and the light source **140** as

the shaft **126** and the light source **140** telescopically move relative to the housing **124**.

(53) As noted above, the electrosurgical device **112** can additionally include features that provide for evacuating surgical smoke from a target tissue to a location external to the surgical site. Surgical smoke is a by-product of various surgical procedures. For example, during surgical procedures, surgical smoke may be generated as a by-product of electrosurgical units (ESU), lasers, electrocautery devices, ultrasonic devices, and/or other powered surgical instruments (e.g., bones saws and/or drills). In some instances, the surgical smoke may contain toxic gases and/or biological products that result from a destruction of tissue. Additionally, the surgical smoke may contain an unpleasant odor. For these and other reasons, many guidelines indicate that exposure of surgical personnel to surgical smoke should be reduced or minimized.

(54) To reduce (or minimize) exposure to surgical smoke, a smoke evacuation system may be used during the surgical procedure. In general, the smoke evacuation system may include a suction pump **146** that can generate sufficient suction and/or vacuum pressure to draw the surgical smoke away from the surgical site. In some implementations, the smoke evacuation system may be coupled to an exhaust system (e.g., an in-wall exhaust system) that exhausts the surgical smoke out of an operating room. In other implementations, the smoke evacuation system may filter air containing the surgical smoke and return the air to the operating room. Within examples, the suction pump **146** and the electrosurgical generator **110** can be provided as separate devices or integrated in a single device (e.g., in a common housing).

(55) As shown in FIG. 1, the shaft **126** can include a smoke evacuation channel **148** at a distal end of the shaft **126**. In an example, the smoke evacuation channel **148** can extend circumferentially around the optical waveguide **142** at the distal end of the shaft **126**. The smoke evacuation channel **148** can also include a smoke inlet that circumferentially surrounds the optical waveguide **142** at the distal end of the shaft **126**. In this arrangement, the smoke inlet of the smoke evacuation channel can help to receive surgical smoke into the smoke evacuation channel **148** in all rotational alignments of the shaft **126** relative to the housing **124** and/or the electrosurgical device **112** relative to the target tissue. However, in another example, the smoke evacuation channel **148** can include one or more smoke inlets that do not extend circumferentially around the optical waveguide **142** and/or the electrosurgical electrode **128**.

(56) In some implementations, the smoke evacuation channel **148** and the optical waveguide **142** can be coaxial. For instance, the smoke evacuation channel **148** and the optical waveguide **142** can each have a longitudinal axis that is aligned with a central axis of the shaft **126**. In other implementations, the smoke evacuation channel **148** and the optical waveguide **142** can have respective longitudinal axes that are offset relative to each such that the smoke evacuation channel **148** and the optical waveguide **142** are not coaxial.

(57) In an example, the smoke evacuation channel **148** can include an outer tube that is separated from the optical waveguide **142** by an air gap. For instance, the shaft **126** can include a plurality of standoffs that extend between the optical waveguide **142** and the outer tube of the smoke evacuation channel **148** to provide the air gap between the outer tube and the optical waveguide **142**. In one implementation, the optical waveguide **142** can include the standoffs such that the optical waveguide **142** and the standoffs are formed as a single, monolithic structure. In another implementation, the standoffs can be formed as a single, monolithic structure with the outer tube of the smoke evacuation channel **148**. In another implementation, the standoffs can be separate from the outer tube of the smoke evacuation channel **148** and the optical waveguide **142**.

(58) In an example, the smoke evacuation channel **148** of the shaft **126** defines a first portion of a smoke flow path, and the interior bore **125** of the housing **124** defines a second portion of a smoke flow path. FIG. 2 illustrates a partial cross-sectional view of the electrosurgical device **112** according to an implementation of this example. In this arrangement, the surgical smoke can be received from the surgical site into the smoke evacuation channel **148** of the shaft **126**, and flow proximally along the smoke evacuation channel **148** to the interior bore **125** of the housing **124**. In

the interior bore **125** of the housing **124**, the smoke can further flow to a smoke tube **150** that is coupled to a proximal end of the housing **124** and configured to convey smoke from the housing **124** to the suction pump **146**.

(59) In another example, the housing **124** includes an interior wall separating the interior bore **125** from a smoke evacuation chamber **152** in the housing **124**. The smoke evacuation channel **148** of the shaft **126** is in fluid communication with the smoke evacuation chamber **152** of the housing **124**. In this example, the smoke evacuation channel **148** of the shaft **126** defines a first portion of a smoke flow path, and the smoke evacuation chamber **152** of the housing **124** defines a second portion of a smoke flow path. FIG. 3 illustrates a partial cross-sectional view of the electrosurgical device **112** according to an implementation of this example. Accordingly, in this example, the smoke is routed through the smoke evacuation chamber **152**, which is separate from the interior bore **125**, to the smoke tube **150** at the proximal end of the housing **124**. This can beneficially help to mitigate exposing one or more components of the electrosurgical device **112** to the surgical smoke in the housing **124**. In some implementations, providing a separate smoke evacuation chamber **152** can additionally or alternatively help to improve the flow of surgical smoke by reducing (or eliminating) obstacles and impediments to gas flows along the second portion of the flow path.

(60) In one implementation, a proximal portion of the smoke evacuation channel **148** comprises at least one aperture **354**, and the interior wall of the housing **124** includes at least one slot. The at least one aperture **354** of the smoke evacuation channel **148** can be aligned with the at least one slot **358** of the interior wall **356** of the housing **124** such that the smoke evacuation channel **148** of the shaft **126** is in fluid communication with the smoke evacuation chamber **152** of the housing **124**. The at least one aperture **354** is axially movable along the at least one slot **358** when the shaft **126** telescopically moves relative to the housing **124** such that the smoke evacuation channel **148** of the shaft **126** is in fluid communication with the smoke evacuation chamber **152** of the housing **124** when the shaft **126** telescopically moves relative to the housing **124**.

(61) In an example, the at least one aperture **354** includes a plurality of apertures **354** and the at least one slot **358** includes a plurality of slots **358**. Also, in this example, each apertures **354** is aligned with a respective one of the plurality of slots **358**, and the shaft **126** is rotatable relative to the housing **124**, and the plurality of apertures **354** and the plurality of slots **358** are arranged around a circumference of the shaft **126** such that fluid communication between the smoke evacuation channel **148** and the smoke evacuation chamber **152** is maintained when the shaft **126** is rotated relative to the housing **124**. In one implementation, the interior wall **356** and the slot(s) **358** can rotate together with the shaft **126** and the aperture(s) **354**. By providing a plurality of apertures **354** and respective slots **358**, at least one pair of aperture **354** and slot **358** can be rotationally aligned with the smoke evacuation chamber **152** and thereby provide fluid communication between the smoke evacuation channel **148** and the smoke evacuation chamber **152**.

(62) Referring now to FIG. 4, a perspective view of an implementation of the electrosurgical device **112** is shown according to an example. As shown in FIG. 4, the electrosurgical device **112** includes the housing **124** defining the interior bore **125**, the shaft **126** telescopically moveable in the interior bore **125** of the housing **124**, and the electrosurgical electrode **128** coupled to the shaft **126**. However, as described above, the shaft **126** can be fixedly coupled to the housing **124** such that the shaft **126** is not moveable relative to the housing **124** in other examples.

(63) Additionally, in FIG. 4, the optical waveguide **142** is at a distal end **460** of the shaft **126**. In this arrangement, the optical waveguide **142** can telescopically move with the shaft **126** relative to the housing **124**. In FIG. 4, the optical waveguide **142** extends around the electrosurgical electrode **128**. This can help to emit the light in a relatively uniform manner by reducing (or preventing) shadows due to an orientation of the optical waveguide **142** and the electrosurgical electrode **128** relative to the surgical site. However, in other examples, the optical waveguide **142** may not extend entirely around the electrosurgical electrode **128** at the distal end **460** of the shaft **126**, and/or the

optical waveguide **142** can be at a different position on the shaft **126** and/or the housing **124**.

(64) In some examples, the electrosurgical device **112** can include a collar **462** at a proximal end of the housing **124**. The collar **462** can be rotatable relative to the housing **124** to increase and/or decrease friction between an outer surface of the shaft **126** and an inner surface of the collar **462**. In this way, the collar **462** to allow and/or inhibit axial telescopic movement of the shaft **126** relative to the housing **124**.

(65) As shown in FIG. 4, the electrosurgical device **112** includes the power cord **122**. At a proximal end **464** of the power cord **122**, the power cord **122** includes a plug **466** configured to couple to the connector **120** of the electrosurgical generator **110**. A distal end of the power cord **122** is coupled to the printed circuit board **132** in an interior cavity of the housing **124**. In this arrangement, the power cord **122** extends proximally from the housing **124** to the plug **466**.

(66) Additionally, as shown in FIG. 4, the user input device(s) **130** include a first button **430A**, a second button **430B**, and a third button **430C** on an exterior surface of the housing **124**. In one implementation, the first button **430A** can be actuated to operate the electrosurgical device **112** in a cutting mode of operation, the second button **430B** can be actuated to operate the electrosurgical device **112** in a coagulation mode of operation, and the third button **430C** can be actuated to operate the light source **140** (i.e., to cause the light source **140** to emit light or cease emitting light). As described above, the user input device(s) **130** can be configured differently in other examples. For instance, the electrosurgical device **112** can be operable in a lesser quantity of modes of operation, a greater quantity of modes of operation, and/or different types of modes of operation in other examples (e.g., such as the example modes of operation described above). Additionally, for instance, the at least one user input device **130** can additionally or alternatively include the user interface **116** of the electrosurgical generator **110** and/or another external device (e.g., a footswitch) for operating the electrosurgical device **112** in one or more modes of operation.

(67) FIGS. 5-6 illustrate a distal end of the electrosurgical device **112** according to examples. As shown in FIGS. 5-6, the shaft **126** includes the optical waveguide **142** at a distal end of the shaft **126**, and the smoke evacuation channel **148** circumferentially surrounding the optical waveguide **142** at the distal end of the shaft **126**. As shown in FIG. 6, the smoke evacuation channel **148** can include an outer tube **670** that is separated from the optical waveguide **142** by a plurality of standoffs **672**.

(68) Referring now to FIGS. 7A-7C, an implementation of the electrosurgical device **112** is shown according to another example. In particular, FIG. 7A shows a perspective view of the electrosurgical device **112**, FIG. 7B shows a perspective view of a distal portion of the electrosurgical device **112**, and FIG. 7C shows a side view of the electrosurgical device **112** according to this example.

(69) As shown in FIG. 7A, the electrosurgical device **112** includes the housing **124** defining the interior bore **125**, the shaft **126** telescopically moveable in the interior bore **125** of the housing **124**, and the electrosurgical electrode **128** coupled to the shaft **126**. The shaft **126** has a longitudinal axis **768** extending from a proximal end **726A** of the shaft **126** to a distal end **726B** of the shaft **126**. Although the shaft **126** is telescopically moveable relative to the housing **124** in FIGS. 7A-7C, the shaft **126** can be fixedly coupled to the housing **124** such that the shaft **126** is not moveable relative to the housing **124** in other examples.

(70) As shown in FIGS. 7B-7C, the shaft **126** includes a shaft wall **770** having an inner surface **770A** and an outer surface **770B**. The inner surface **770A** of the shaft wall **770** defines the smoke evacuation channel **148** that extends from the distal end **726B** of the shaft **126** to the proximal end **726A** of the shaft **126**. As described above, the smoke evacuation channel **148** can provide the first portion of the smoke flow path along which surgical smoke flows as the surgical smoke is evacuated from a surgical site by the electrosurgical device **112**.

(71) For instance, in one example, the smoke evacuation channel **148** of the shaft **126** can define the first portion of the smoke flow path, and the interior bore **125** of the housing **124** can define the

second portion of a smoke flow path as described above with respect to FIG. 2. In this example, the smoke received from the surgical site into the smoke evacuation channel **148** of the shaft **126** can flow proximally along the smoke evacuation channel **148** to the interior bore **125** of the housing **124**. In the interior bore **125** of the housing **124**, the smoke can further flow to the smoke tube **150** (shown in FIG. 1) that is coupled to a proximal end of the housing **124** and configured to convey the smoke from the housing **124** to the suction pump **146** (shown in FIG. 1).

(72) In another example, the smoke evacuation channel **148** of the shaft **126** can define the first portion of the smoke flow path, and the smoke evacuation chamber **152** of the housing **124** can define the second portion of the smoke flow path as described above with respect to FIG. 3. In this example, as described above, the smoke can be routed through the smoke evacuation chamber **152** (shown in FIGS. 1 and 3), which is separate from the interior bore **125**, to the smoke tube **150** at the proximal end of the housing **124**. This can beneficially help to mitigate exposing one or more components of the electrosurgical device **112** to the surgical smoke in the housing **124**. In some implementations, providing a separate smoke evacuation chamber **152** can additionally or alternatively help to improve the flow of surgical smoke by reducing (or eliminating) obstacles and impediments to gas flows along the second portion of the flow path.

(73) As noted above, the electrosurgical electrode **128** is coupled to the shaft **126**. More specifically, in FIGS. 7A-7C, the electrosurgical device **112** includes an electrode receptacle **737** that couples the electrosurgical electrode **128** to the shaft **126**. As shown in FIGS. 7B-7C, the electrode receptacle **737** extends across the smoke evacuation channel **148** in a direction that is transverse to the longitudinal axis **768** of the shaft **126** such that electrode receptacle **737** partitions the smoke evacuation channel **148** into a plurality of smoke inlets **772** at the distal end **726B** of the shaft **126**. As such, the smoke inlets **772** can be located on a plurality of sides of the electrode receptacle **737** and the electrosurgical electrode **128** (e.g., a first smoke inlet **772** can be adjacent a first major surface of the electrosurgical electrode **128** and a second smoke inlet **772** can be adjacent a second, opposing major surface of the electrosurgical electrode **128**). In this arrangement, the smoke inlets **772** of the smoke evacuation channel **148** can help to receive surgical smoke into the smoke evacuation channel **148** in all rotational alignments of the shaft **126** relative to the housing **124** and/or the electrosurgical device **112** relative to the target tissue.

(74) In one example, a cross-sectional area of the smoke inlets **772** can be at least 50% of a cross-sectional area of the smoke evacuation channel **148**. In another example, the cross-sectional area of the smoke inlets **772** can be at least 75% of the cross-sectional area of the smoke evacuation channel **148**. In another example, the cross-sectional area of the smoke inlets **772** can be at least 80% of the cross-sectional area of the smoke evacuation channel **148**. In another example, the cross-sectional area of the smoke inlets **772** can be at least 85% of the cross-sectional area of the smoke evacuation channel **148**. Accordingly, the electrode receptacle **737** can help to achieve a relatively large cross-sectional area of the smoke inlets **772** (and the smoke evacuation channel **148**), which can help to increase smoke evacuation performance of the electrosurgical device **112**.

(75) In FIG. 7C, a first one of the smoke inlets **772** is defined by a first side of the electrode receptacle **737** and a first portion of the inner surface **770A** of the shaft wall **770**, and a second one of the smoke inlets **772** is defined by a second side of the electrode receptacle **737** and a second portion of the inner surface **770A** of the shaft wall **770**. In some examples, the smoke inlets **772** can have a common size and/or a common shape. This can help to provide relatively balanced suction on the first side of the electrode receptacle **737** and the second side of the electrode receptacle **737**. However, in other examples, at least one of the smoke inlets **772** can have a different size and/or a different shape than another one of the smoke inlets **772**.

(76) Within examples, the electrode receptacle **737** can define an aperture **774** that can receive the electrosurgical electrode **128**, and the electrosurgical electrode **128** can be coupled to the electrode receptacle **737** by a releasable coupling (e.g., friction-fit or a threaded coupling) or a non-releasable coupling (e.g., via welding and/or soldering). The aperture **774** can be aligned with a center axis of

the shaft **126** such that the electrosurgical electrode **128** is centered on the distal end **726B** of the shaft **126**. This can help to provide for relatively good visibility of the electrosurgical electrode **128** and the surgical site. Additionally, the center location of the electrosurgical electrode **128** can help to provide for operating the electrosurgical device **112** in a plurality of rotational alignments between the electrosurgical electrode **128** relative to the housing **124** and/or between the electrosurgical device **112** relative to the surgical site.

(77) As noted above, in some examples, the electrosurgical device **112** can include one or more optical fibers in addition or alternative to the optical waveguide **142** for transmitting the light distally from the light source **140** to the surgical field. FIGS. 7A-7C depict an example in which the electrosurgical device **112** includes a plurality of optical fibers **776** extending from the proximal end **726A** of the shaft **126** to the distal end **726B** of the shaft **126**. In the example of FIGS. 7A-7C, the electrosurgical device **112** omits the optical waveguide **142**. However, the electrosurgical device **112** shown in FIGS. 7A-7C can include the optical waveguide **142** in addition or alternative to the optical fibers **776** in another example.

(78) As shown in FIG. 7A, the electrosurgical device **112** includes the light source **140** coupled to the plurality of optical fibers **776** at a position that is proximal of the proximal end **726A** of the shaft **126**. In one example, the light source **140** can be in the housing **124** proximal of the shaft **126**. In another example, as shown in FIG. 7A, the electrosurgical device **112** includes the power cord **122** extending proximally from the housing **124** to a plug **766**, and the light source **140** is located in a housing of the plug **766**. This can help to reduce (or eliminate) elevated temperatures in the housing **124** and/or the shaft **126** of the electrosurgical device **112**. In an implementation, each optical fiber **776** can extend from the light source **140** in the plug **766** to the distal end **726B** of the shaft **126**. In this implementation, the optical fibers **776** can extend through the power cord **122**, the interior bore **125** of the housing **124**, and the smoke evacuation channel **148** of the shaft **126**.

(79) At the distal end **726B** of the shaft **126**, the optical fibers **776** can be coupled to a plurality of conduits **778**, which can provide for fixedly positioning light-emitting ends of the optical fibers **776** relative to the distal end **726B** of the shaft **126** and/or the electrosurgical electrode **128**. The optical fibers **776** can be coupled to the respective conduits **778** by, for instance, a friction-fit coupling, an adhesive coupling, and/or a welded coupling.

(80) As shown in FIG. 7C, the conduits **778** can extend through the shaft wall **770** between the inner surface **770A** and the outer surface **770B**, and each optical fiber **776** can be in a respective one of the conduits **778**. This can provide for positioning the light-emitting ends of the optical fibers **776** at a periphery of the shaft **126**, which can help to reduce (or minimize) the formation of shadows on the surgical site. In some examples, the light-emitting ends of the optical fibers **776** can be arranged in a plane that is approximately perpendicular to the longitudinal axis **768** (e.g., parallel with a plane of the distal end **726B** of the shaft **126**). In other examples, one or more of the light-emitting ends of the optical fibers **776** can be arranged at a different angle relative to the longitudinal axis **768** (e.g., angled inwardly toward the longitudinal axis **768** and/or angled outwardly away from the longitudinal axis **768**).

(81) In FIGS. 7A-7C, a quantity of the conduits **778** is four, and a quantity of the optical fibers **776** is four. However, the electrosurgical device **112** can include a lesser quantity or a greater quantity of the conduits **778** and the optical fibers **776** in other examples. In some examples, the quantity of the conduits **778** and the optical fibers **776** can be at least two. This can help to illuminate the surgical field around the electrosurgical electrode **128** and thereby mitigate (or eliminate) the formation of shadows on the surgical field. In FIGS. 7A-7C, the conduits **778** and the optical fibers **776** are equally spaced from each other around the periphery of the shaft **126**. This can also help to illuminate the surgical field around the electrosurgical electrode **128** and thereby mitigate (or eliminate) the formation of shadows on the surgical field.

(82) In some implementations, the light-emitting ends of the optical fibers **776** can include one or more optical structures that can help to shape the light emitted from the optical fibers **776**. As

examples, the one or more optical structures can be selected from a group consisting of: one or more lenses, one or more prisms, one or more facets, and one or more optical filters.

(83) In one implementation, the electrode receptacle **737** and a remainder of the shaft **126** can be integrally formed as a single-piece, monolithic structure. In another implementation, the shaft **126** can be formed as multi-part construction. For instance, in FIGS. 7A-7C, the shaft **126** can include a distal end portion **780** coupled to a main body portion **782**, and the distal end portion **780** can include the electrode receptacle **737**. In this example, the distal end portion **780** is press-fit into the main body portion **782** of the shaft **126**. In other examples, the distal end portion **780** can be additionally or alternatively coupled to the main body portion **782** by an adhesive, bonding, welding, and/or soldering.

(84) Within examples, the distal end portion **780** and the main body portion **782** can include one or more alignment features that facilitate coupling the distal end portion **780** and the main body portion **782** in a predetermined rotational alignment relative to each other. For instance, in FIG. 7B, the distal end portion **780** includes a protrusion **784A** and the main body portion **782** includes a slot **784B** that receives the protrusion **784A** when the distal end portion **780** is coupled to the main body portion **782** of the shaft **126**. The alignment features of the distal end portion **780** of the shaft **126** and the main body portion **782** of the shaft **126** can help to arrange the electrosurgical electrode **128** is a predetermined orientation relative to other features of the electrosurgical device **112** such as, for instance, the user input device(s) **130**.

(85) In FIGS. 7A-7C, the distal end portion **780** of the shaft **126** includes the conduits **778** extending through the shaft wall **770** between the inner surface **770A** and the outer surface **770B**. In an implementation, the main body portion **782** can omit the conduits **778**. In this implementation, each optical fiber **776** can be fixedly coupled to the shaft wall **770** by a respective one of the conduits **778** at the distal end portion **780** and can be unsecured to the shaft wall **770** in the main body portion **782** of the shaft **126**. In another implementation, the main body portion **782** can additionally include the conduits **778** to couple the optical fibers **776** to the shaft wall **770** in the main body portion **782**.

(86) In the example described above and shown in FIGS. 7A-7C, the optical fibers **776** are coupled to the shaft **126** by the conduits **778**. In other examples, the optical fibers **776** can be additionally or alternatively coupled to shaft **126** by other optical fiber retainer structures such as, for instance, one or more clips and/or one or more grooves in which the optical fibers can be retained by a friction-fit coupling.

(87) In the example shown in FIGS. 7A-7C, the electrosurgical device **112** includes four conduits **778** and four optical fibers **776** equally spaced from each other around the periphery of the shaft **126**. As described above, the electrosurgical device **112** can include a lesser quantity or a greater quantity of the conduits **778** and the optical fibers **776** in other examples. For instance, FIGS. 8A-8B depict a distal end portion **880** of the shaft **126** that includes two conduits **878A**, **878B** that can couple two optical fibers **776** to the shaft **126**, according to another example.

(88) As shown in FIGS. 8A-8B, the conduits **878A**, **878B** can include a first conduit **878A** and a second conduit **878B**. In this example, an electrode receptacle **837** extends across the smoke evacuation channel **148** between the first conduit **878A** and the second conduit **878B**. In this arrangement, the distal end portion **880** is configured to couple a first optical fiber and a second optical fiber on opposing sides of the electrosurgical electrode **128** when the electrosurgical electrode **128** is disposed in an aperture **874** of the electrode receptacle **837**. This can help to emit light on a plurality of sides of the electrosurgical electrode **128** and thereby reduce (or prevent) forming shadows on the surgical site.

(89) Additionally, as shown in FIG. 8A, the distal end portion **880** of the shaft **126** can include a proximal-extension **886** that extends proximally from the distal end **726B** of the shaft **126**. Within examples, the distal end portion **780** shown in FIGS. 7A-7C can also include or omit the proximal-extension **886**. The proximal-extension **886** can be sized and/or shaped such that the proximal-

extension **886** can be received into the smoke evacuation channel **148** of the main body portion **782** when the distal end portion **880** is coupled to the main body portion **782**. For instance, the proximal-extension **886** can have dimensions that are relatively smaller than dimensions of the shaft wall **770** at the main body portion **782** to allow the main body portion **882** to receive the proximal-extension **886** of the distal end portion **880**. The proximal-extension **886** received in the main body portion **782** can help to provide for a friction-fit coupling between the distal end portion **880** and the main body portion **782**, and/or improve a strength of the coupling between the distal end portion **880** and the main body portion **782**.

(90) Referring now to FIGS. **9A-9E**, an implementation of the electrosurgical device **112** is shown according to another example. In particular, FIG. **9A** shows a perspective view of the electrosurgical device **112**, FIG. **9B** shows a perspective view of a distal end **926B** of the shaft **126** of the electrosurgical device **112**, FIG. **9C** shows a side view of the distal end **926B** of the shaft **126**, FIG. **9D** shows a cross-sectional view of the electrosurgical device **112** through a center axis shown in FIG. **9A**, and FIG. **9E** shows an exploded view of the electrosurgical device **112**.

(91) As shown in FIG. **9A**, the electrosurgical device **112** includes the housing **124** defining the interior bore **125**, the shaft **126** telescopically moveable in the interior bore **125** of the housing **124**, and the electrosurgical electrode **128** coupled to the shaft **126**. The shaft **126** has a center axis **968** extending from a proximal end of the shaft **126** to the distal end **926B** of the shaft **126**. Although the shaft **126** is telescopically moveable relative to the housing **124** in FIGS. **9A-9C**, the shaft **126** can be fixedly coupled to the housing **124** such that the shaft **126** is not moveable relative to the housing **124** in other examples.

(92) As shown in FIGS. **9B-9C**, the shaft **126** includes a shaft wall **970** having an inner surface **970A** and an outer surface **970B**. The inner surface **970A** of the shaft wall **970** defines the smoke evacuation channel **148** that extends from the distal end **926B** of the shaft **126** to the proximal end of the shaft **126**. As described above, the smoke evacuation channel **148** can provide the first portion of the smoke flow path along which surgical smoke flows as the surgical smoke is evacuated from a surgical site by the electrosurgical device **112**.

(93) For instance, in an example shown in FIG. **9C**, the smoke evacuation channel **148** of the shaft **126** can define the first portion of the smoke flow path, and the interior bore **125** of the housing **124** can define the second portion of a smoke flow path as described above with respect to FIG. **2**. In this example, the smoke received from the surgical site into the smoke evacuation channel **148** of the shaft **126** can flow proximally along the smoke evacuation channel **148** to the interior bore **125** of the housing **124**. In the interior bore **125** of the housing **124**, the smoke can further flow to the smoke tube **150** (shown in FIG. **1**) that is coupled to a proximal end of the housing **124** and configured to convey the smoke from the housing **124** to the suction pump **146** (shown in FIG. **1**).

(94) In another example, the smoke evacuation channel **148** of the shaft **126** can define the first portion of the smoke flow path, and the smoke evacuation chamber **152** of the housing **124** can define the second portion of the smoke flow path as described above with respect to FIG. **3**. In this example, as described above, the smoke can be routed through the smoke evacuation chamber **152** (shown in FIGS. **1** and **3**), which is separate from the interior bore **125**, to the smoke tube **150** at the proximal end of the housing **124**. This can beneficially help to mitigate exposing one or more components of the electrosurgical device **112** to the surgical smoke in the housing **124**. In some implementations, providing a separate smoke evacuation chamber **152** can additionally or alternatively help to improve the flow of surgical smoke by reducing (or eliminating) obstacles and impediments to gas flows along the second portion of the flow path.

(95) As noted above, the electrosurgical electrode **128** is coupled to the shaft **126**. As shown in FIGS. **9A** and **9C-9E**, a distal portion of the electrosurgical electrode **128** extends distally from the shaft **126**, and the distal portion of the electrosurgical electrode **128** extends inwardly from a periphery of the shaft **126** to the center axis **968** of the shaft **126**. The inwardly extending shape of the distal portion of the electrosurgical electrode **128** can help to provide greater flexibility in

positioning and arranging the smoke evacuation channel **148** and/or an illumination feature (e.g., the light source **140**, the optical waveguide **142**, and/or the optical fibers **770**) at the distal end **926B** of the shaft **126**.

(96) In FIGS. **9A-9E**, the electrosurgical electrode **128** includes (i) a first electrode portion **928A** coupled to the periphery of the shaft **126** and extending along a direction that is parallel to the center axis **968**, (ii) a second electrode portion **928B** extending from the first electrode portion **928A** to the center axis **968**, and (iii) a third electrode portion **928C** extending distally from the second electrode portion **928B** and along the center axis **968** of the shaft **926**. The electrosurgical electrode **128** can also include a first bend **928D** between the first electrode portion **928A** and the second electrode portion **928B**, and a second bend **928E** between the second electrode portion **928B** and the third electrode portion **928C**.

(97) In an example, an angle between the first electrode portion **928A** and the second electrode portion **928B** is less than approximately 90 degrees. This can help to improve the flow of surgical smoke into the smoke evacuation channel **148** relative to an alternative implementation in which the angle between the first electrode portion **928A** and the second electrode portion **928** is 90 degrees.

(98) In operation, the surgical smoke can flow along a smoke flow path **988** shown in FIG. **9D**. For instance, as shown in FIG. **9D**, the surgical smoke can flow (i) along the second electrode portion **928B** to enter the smoke evacuation channel **148** of the shaft **126**, (ii) from the smoke evacuation channel **148** to the interior bore **125** of the housing **124**, and (iii) from the interior bore **125** of the housing **124** to the smoke tube **150** at the proximal end of the housing **124**.

(99) As shown in FIGS. **9C** and **9E**, the shaft **126** can include (i) a first shaft portion **926A** that extends between the proximal end and the distal end, and (ii) a second shaft portion **990B** that extends between the proximal end and the distal end. As shown in FIG. **9C**, the smoke evacuation channel **148** can be defined between the first shaft portion **990A** and the second shaft portion **990B**. In this example, the electrosurgical electrode **128** is coupled to the second shaft portion **990B**. Although FIGS. **9A-9E** show the shaft **126** including the first shaft portion **990A** and the second shaft portion **990B**, which can be coupled to each other, the shaft **126** can be a single-part, monolithic structure that defines the smoke evacuation channel **148** in other examples.

(100) As shown in FIG. **9E**, the electrosurgical device **112** can also include the electrical components that facilitate supplying the electrosurgical energy, which the electrosurgical device **112** receives from the electrosurgical generator **110**, to the electrosurgical electrode **128**. For example, as described above with respect to FIG. **1**, the electrosurgical device **112** shown in FIG. **9E** can include the printed circuit board **132**, the housing conductor **134**, and the conductive lead(s) **136** to form a circuit for conducting the electrosurgical energy from the power cord **122** to the electrosurgical electrode **128**. In FIG. **9E**, the housing conductor **134** is in the form of a conductive channel on an interior surface of the housing **124**, and the conductive lead **136** is in the form of a conductive track that extends along the inner surface **970A** of the shaft **126** with a portion of the conductive lead **136** being at the outer surface **970B** of the shaft **126** at the proximal end of the shaft **126**. In this arrangement, a distal end of the conductive lead **136** can electrically couple to the electrosurgical electrode **128** and a proximal end of the conductive lead **136** can electrically couple to the housing conductor **134** as the shaft **126** telescopically moves relative to the housing **124**.

(101) As noted above, the electrosurgical device **112** can include an illumination feature at the distal end **926B** of the shaft **126**. For example, as described above, the electrosurgical device **112** can include an illumination feature selected from a group consisting of one or more light sources, one or more optical fibers, and one or more optical waveguides. In the example shown in FIGS. **9B-9C**, the electrosurgical device **112** includes the optical waveguide **142** extending through a passage **992** in the first shaft portion **990A**. However, in another example, the electrosurgical device **112** can include the light source **140** at the distal end **926B** of the shaft **126** and/or one or more optical fibers (e.g., the optical fibers **776**) at the distal end **926B** of the shaft **126**.

(102) FIGS. **10A-10E** depicts an implementation of the electrosurgical device **112** in which the electrosurgical electrode **128** extends inwardly from a periphery of the shaft **126** to the center axis of the shaft **126** according to another example. In particular, FIG. **10A** depicts a perspective view of the electrosurgical device **112**, FIG. **10B** depicts a perspective view of a distal end of the shaft **126** of the electrosurgical device **112** shown in FIG. **10A**, FIG. **10C** shows a side view of a distal end of the shaft **126** of the electrosurgical device **112** shown in FIG. **10A**, FIG. **10D** shows another side view of a distal end of the shaft **126** of the electrosurgical device **112** shown in FIG. **10A**, and FIG. **10E** shows a ring structure of the electrosurgical device **112** shown in FIG. **10A**.

(103) As shown in FIGS. **10A-10D**, the electrosurgical device **112** includes the housing **124** defining the interior bore **125**, the shaft **126** telescopically moveable in the interior bore **125** of the housing **124**, and the electrosurgical electrode **128** coupled to the shaft **126**. The shaft **126** has a center axis **1068** extending from a proximal end of the shaft **126** to the distal end **1026B** of the shaft **126**. Although the shaft **126** is telescopically moveable relative to the housing **124** in FIGS. **10A-10-D**, the shaft **126** can be fixedly coupled to the housing **124** such that the shaft **126** is not moveable relative to the housing **124** in other examples.

(104) In FIGS. **10A-10D**, a distal portion of the electrosurgical electrode **128** extends distally from the shaft **126**, and the distal portion of the electrosurgical electrode **128** extends inwardly from a periphery of the shaft **126** to the center axis **1068** of the shaft **126**. The inwardly extending shape of the distal portion of the electrosurgical electrode **128** can help to provide greater flexibility in positioning and arranging the smoke evacuation channel **148** and/or an illumination feature (e.g., a light source **140**) at the distal end **1026B** of the shaft **126**.

(105) As shown in FIGS. **10D-10E**, the electrosurgical device **112** can include a light-emitting diode printed circuit board (LED PCB) **1092** in the smoke evacuation channel **148** of the shaft **126**. The LED PCB **1092** can include an aperture **1094** having a diameter that is smaller than a diameter of the smoke evacuation channel **148**. The aperture **1094** can allow for smoke to pass through the LED PCB **1092** while evacuating smoke from the surgical site.

(106) Additionally, as shown in FIGS. **10D-10E**, the LED PCB **1092** can also include one or more electrical contacts **1095** electrically coupled to one or more LEDs **1096**. The one or more electrical contacts **1095** can also couple to the conductive lead(s) **136** (shown in FIG. **1**). In this way, the conductive lead(s) **136** can supply electrical power to the LEDs **1096** via the one or more electrical contacts **1095**. In some examples, LED PCB **1092** can be plated to help reduce RF interference to the one or more LEDs **1096**. FIG. **11** depicts a plurality of example LED PCBs **1192A-1192G** that can be used with the electrosurgical device **112** shown in FIGS. **10A-10E** according to examples.

(107) FIG. **10F** shows a cross-sectional view of the electrosurgical device **112** through the longitudinal axis **1068**, according to an example. As shown in FIG. **10F**, the electrosurgical electrode **128** is integrally formed as a single-piece, monolithic structure with the conductive lead **136**. In an example, the conductive lead **136** can have a semi-circle shape that corresponds to a circular shape of the inner surface of the shaft **126**. The conductive lead **136** extends proximally from the electrosurgical electrode **128** to a conductive spring **1039**. The conductive spring **1039** can extend around and/or through the shaft wall of the shaft **126** to electrically couple conductive lead **136** (and the electrosurgical electrode **128**) to the housing conductor **134** of the housing **124**. In this way, the conductive spring **1039** can provide for maintaining an electrical coupling between the conductive lead **136** and the housing conductor **134** in all axial positions of the shaft **126** relative to the housing **124**.

(108) The description of the different advantageous arrangements has been presented for purposes of illustration and description, and is not intended to be exhaustive or limited to the examples in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art. Further, different advantageous examples may describe different advantages as compared to other advantageous examples. The example or examples selected are chosen and described in order to explain the principles of the examples, the practical application, and to enable others of ordinary

skill in the art to understand the disclosure for various examples with various modifications as are suited to the particular use contemplated.

Claims

1. An electrosurgical device, comprising: a housing defining an interior bore; a shaft telescopically moveable in the interior bore of the housing, wherein the shaft comprises: an optical waveguide at a distal end of the shaft, and a smoke evacuation channel circumferentially surrounding the optical waveguide at the distal end of the shaft; and an electrosurgical electrode coupled to the shaft.
2. The electrosurgical device of claim 1, wherein the smoke evacuation channel and the optical waveguide are coaxial.
3. The electrosurgical device of claim 1, further comprising a plurality of standoffs separating an outer tube of the smoke evacuation channel from the optical waveguide.
4. The electrosurgical device of claim 1, wherein the smoke evacuation channel of the shaft defines a first portion of a smoke flow path, and wherein the interior bore of the housing defines a second portion of the smoke flow path.
5. The electrosurgical device of claim 4, further comprising a smoke tube coupled to a proximal end of the housing and configured to convey smoke from the housing to a suction pump.
6. The electrosurgical device of claim 1, wherein the housing comprises an interior wall separating the interior bore from a smoke evacuation chamber in the housing, wherein the smoke evacuation channel of the shaft is in fluid communication with the smoke evacuation chamber of the housing, wherein the smoke evacuation channel of the shaft defines a first portion of a smoke flow path, and wherein the smoke evacuation chamber of the housing defines a second portion of the smoke flow path.
7. The electrosurgical device of claim 6, wherein a proximal portion of the smoke evacuation channel comprises at least one aperture, wherein the interior wall of the housing comprises at least one slot, and wherein the at least one aperture of the smoke evacuation channel is aligned with the at least one slot of the interior wall of the housing such that wherein the smoke evacuation channel of the shaft is in fluid communication with the smoke evacuation chamber of the housing.
8. The electrosurgical device of claim 7, wherein the at least one aperture comprises a plurality of apertures and the at least one slot comprises a plurality of slots, wherein each aperture is aligned with a respective one of the plurality of slots, wherein the shaft is rotatable relative to the housing, and wherein the plurality of apertures and the plurality of slots are arranged around a circumference of the shaft such that fluid communication between the smoke evacuation channel and the smoke evacuation chamber is maintained when the shaft is rotated relative to the housing.
9. The electrosurgical device of claim 7, wherein the at least one aperture is axially movable along the at least one slot when the shaft telescopically moves relative to the housing.
10. The electrosurgical device of claim 6, further comprising a printed circuit board fixedly coupled to the housing in the interior bore, wherein the printed circuit board comprises a plurality of switches that are operable to control a supply of electrosurgical energy from an electrosurgical generator to the electrosurgical electrode.
11. The electrosurgical device of claim 1, wherein the shaft is rotatable about an axis of rotation that is parallel to a longitudinal axis of the electrosurgical device.
12. The electrosurgical device of claim 1, wherein the electrosurgical electrode is rotatable relative to the shaft.
13. The electrosurgical device of claim 1, further comprising a light source optically coupled to the optical waveguide.
14. The electrosurgical device of claim 13, wherein the light source is coupled to the shaft such that the light source is telescopically movable with the shaft relative to the housing.
15. The electrosurgical device of claim 13, wherein the light source is in the interior bore of the

housing.

16. The electrosurgical device of claim 13, wherein the light source is coupled to an exterior surface of the housing.

17. The electrosurgical device of claim 1, wherein the electrosurgical electrode extends from a central portion of the optical waveguide, and wherein the optical waveguide extends entirely around the electrosurgical electrode.

18. The electrosurgical device of claim 17, wherein the electrosurgical electrode extends from a central portion of the optical waveguide, and wherein the optical waveguide does not extend entirely around the electrosurgical electrode at the distal end of the shaft.

19. A method, comprising: providing the electrosurgical device of claim 1; supplying electrosurgical energy to the electrosurgical electrode; and applying suction to the smoke evacuation channel.

20. The method of claim 19, wherein applying the suction to the smoke evacuation channel is performed while supplying the electrosurgical energy to the electrosurgical electrode.
